

CAMPINA, Romania

WMO: 153490

Lat: 45.1436N

Lon: 25.7333E

Elev: 1517

StdP: 13.91

Time Zone: 2.00 (EUE)

Period: 87-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.1	15.1	3.9	7.1	15.6	7.6	8.6	19.3	26.7	32.9	16.4	34.7	2.0	0	0.458

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.4	86.6	68.0	84.2	67.1	81.6	66.4	70.6	82.6	69.1	80.7	67.7	78.9	4.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.6	103.5	75.4	65.1	98.1	74.0	63.7	93.3	72.6	35.4	83.4	34.1	81.1	32.9	79.2	76.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.1	12.1	9.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.2	30.4	34.7	40.7	49.1	58.7	66.1	68.2	68.4	60.1	49.9	40.7	34.5		
	DBStd	15.02	8.21	8.10	7.89	7.71	5.58	5.05	4.63	5.12	6.59	7.01	7.89	7.76		
	HDD50	2316	607	430	300	110	6	0	0	0	4	87	292	480		
	HDD65	5769	1072	848	754	478	206	46	23	28	173	469	728	944		
	CDD50	2388	0	2	11	82	274	483	563	569	306	83	14	1		
	CDD65	367	0	0	0	0	9	79	121	132	26	0	0	0		
	CDH74	3577	0	0	0	28	149	754	1058	1261	312	15	0	0		
	CDH80	918	0	0	0	1	23	158	274	395	67	0	0	0		
	WSAvg	3.4	2.9	3.4	4.2	4.3	3.7	3.5	3.6	3.1	3.4	3.1	3.2	2.9		
	PrecAvg	31.3	1.6	1.3	1.7	2.6	3.7	4.4	4.0	3.1	2.7	2.4	1.9	2.0		
(15) Precipitation	PrecMax	43.6	3.4	3.4	3.1	6.6	7.8	7.3	8.7	9.6	5.4	5.4	4.1	5.2		
	PrecMin	18.1	0.4	0.2	0.3	1.1	0.8	1.8	1.6	0.8	0.9	0.2	0.1	0.2		
	PrecStd	6.3	0.8	0.7	0.8	1.3	1.7	1.5	1.8	1.8	1.3	1.3	1.1	1.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.6	62.7	69.8	77.4	82.9	87.4	89.8	90.2	85.5	76.1	67.0	61.5		
		MCWB	45.9	50.1	54.2	58.1	64.1	69.7	71.1	68.8	64.1	59.5	54.3	48.9		
	2%	DB	50.4	56.1	64.2	72.6	77.8	84.1	85.9	86.8	81.3	71.9	61.4	54.3		
		MCWB	42.2	45.4	49.8	56.9	61.6	67.7	69.0	67.4	63.9	58.6	51.6	45.1		
	5%	DB	45.6	51.4	59.3	68.5	75.0	81.3	83.2	84.3	77.2	67.4	56.7	49.5		
		MCWB	39.2	43.5	47.9	55.0	60.7	67.1	67.8	66.8	62.4	55.8	50.1	43.3		
	10%	DB	41.2	46.5	54.5	64.0	71.3	78.4	80.4	81.1	73.5	62.6	52.0	45.2		
		MCWB	36.9	40.3	45.2	51.8	59.0	65.7	66.9	66.3	61.0	53.3	47.5	40.7		
	0.4%	WB	46.1	50.9	54.4	59.9	65.4	72.6	72.5	72.5	67.9	61.9	55.7	49.3		
		MCDB	55.9	62.3	67.4	73.7	77.8	83.8	84.9	86.0	82.1	71.7	64.5	56.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	42.8	46.7	51.0	57.8	63.6	69.9	70.7	69.9	65.3	59.1	53.3	47.0		
		MCDB	49.0	54.5	61.9	71.7	74.8	81.6	82.5	82.6	77.1	69.1	59.5	53.5		
	5%	WB	39.7	44.3	48.3	55.6	61.9	68.2	69.2	68.4	63.4	56.9	50.7	43.9		
		MCDB	44.2	50.7	57.5	66.5	71.8	79.1	80.3	80.4	74.5	66.7	55.2	48.8		
	10%	WB	37.3	41.5	45.8	53.5	59.9	66.6	67.6	67.0	61.9	54.6	48.3	41.2		
		MCDB	40.3	46.0	52.8	62.3	69.1	77.2	78.1	78.5	72.3	61.2	51.7	45.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.2	17.0	18.2	19.6	19.3	20.3	20.4	21.6	20.6	18.9	15.2	15.0		
		MCDBR	21.5	25.1	27.4	27.8	25.5	24.2	24.7	26.4	26.7	27.7	22.1	22.2		
		MCWBR	13.5	15.8	14.9	14.1	12.1	11.5	11.0	10.5	11.4	14.6	13.1	13.4		
		MCDBR	19.9	23.8	25.1	25.7	22.3	22.7	22.2	23.2	23.9	26.0	19.1	20.3		
	5% WB	MCWBR	13.0	15.0	14.6	13.3	11.8	11.3	11.0	10.2	10.7	14.5	11.8	12.8		
(40) Clear-Sky Solar Irradiance	taub		0.320	0.352	0.394	0.456	0.432	0.435	0.448	0.466	0.419	0.380	0.352	0.322		
	taud		2.313	2.223	2.171	2.033	2.175	2.206	2.178	2.118	2.214	2.305	2.336	2.336		
	Ebn at Noon		246	258	262	255	266	266	260	250	252	245	232	232		
	Edh at Noon		25	34	41	51	46	44	45	46	39	31	25	23		
(44) All-Sky Solar Radiation	RadAvg		438	699	1059	1413	1741	1880	1910	1717	1252	840	477	360		
	RadStd		53	94	112	150	152	152	147	124	111	104	52	48		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (70 neighbors)	+0.95	N/A	N/A	+1.17	N/A	N/A	N/A	-275	+137	+65

Nomenclature: See separate page